COMP 2711 – Mock Midterm 07

Solutions and marking notes

Practice material – not an official examination

Problem 1 [10 points]

(a) **[3 points]**

p	q	$p \to q$	$q \to p$	$(p \to q) \oplus (q \to p)$	$p \oplus q$
T	T	T	T	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	F	F

The last two columns agree on every assignment, establishing equivalence.
Marking: 1 all rows and implication columns; 1 both XOR columns; 1 correct equivalence conclusion.

(b) **[2 points]** No. For $p = F, q = T$, the original $p \to q$ is T and the converse $q \to p$ is F. The inverse $\neg p \to \neg q$ is also F, as expected because inverse and converse are equivalent.
Marking: 1 separating assignment with original/converse values; 1 correct inverse and value.

(c) **[5 points]**

1. p and q by simplification of $p \wedge q$.
2. $p \vee r$ from p by addition.
3. s from line 2 and the second premise by modus ponens.
4. t from q and the third premise by modus ponens.
5. $s \wedge t$ by conjunction of lines 3 and 4.

Marking: 1 simplifications; 1 addition; 1 each modus ponens; 1 conjunction and rule citations.

Problem 2 [13 points]

(a) **[4 points]** For D, the expansions are $P(-1) \wedge P(0) \wedge P(1) = T \wedge F \wedge T = F$ and $P(-1) \vee P(0) \vee P(1) = T \vee F \vee T = T$. On the empty domain the universal statement is true because there is no counterexample; the existential is false because there is no witness.
Marking: 1 each finite expansion/value; 1 each empty-domain value with reason.

(b) **[4 points]**

$$\exists x \forall y \exists z \big((\neg P(x, y) \wedge \neg Q(z)) \vee \neg R(x, y, z) \big).$$

Each quantifier changes type; negate the conjunction first, then the inner disjunction.

Marking: 2 all quantifier changes; 1 outer De Morgan; 1 inner De Morgan/scopes.

(c) **[5 points]** A renaming is $(\forall u P(u)) \vee (\exists v Q(v))$. For the inference, existential instantiation gives a fresh a with $\neg Q(a)$. Universal instantiation gives $P(a) \to Q(a)$. Modus tollens yields $\neg P(a)$, and existential generalization yields $\exists x \neg P(x)$.
Marking: 1 capture-free renaming; 1 each of fresh witness, universal instantiation, modus tollens and existential generalization.

Problem 3 [12 points]

(a) **[2 points]** The union is $\{0, 1, 2, 3\}$. The intersection is $\varnothing$: $A_0 = \{0, 1\}$ and $A_2 = \{2, 3\}$ already have no common element.
Marking: 1 union; 1 intersection.

(b) **[3 points]** If $g(f(a)) = g(f(b))$, injectivity of g gives $f(a) = f(b)$ and injectivity of f gives $a = b$. Thus the composition is injective. The converse claim about g fails: take $A = \{0\}$, $B = \{0, 1\}$, $C = \{0\}$, $f(0) = 0$, and $g(0) = g(1) = 0$. The composition on the singleton domain is injective but g is not.
Marking: 2 two cancellations and arbitrary inputs; 1 correct counterexample with compatible domains.

(c) **[7 points]**

(i) **F**. For $A = \{0\}$ and $B = \{1\}$, the subset $\{0, 1\}$ belongs only to the left side.

(ii) **T**. A set is a subset of the intersection exactly when it is a subset of both sets.

(iii) **T**. Every finite length has finitely many strings; list by length.

(iv) **T**. Encode each by the finite tuple of one-positions; for zero ones it is a singleton.

(v) **F**. List triples by their coordinate sum; each layer is finite.

(vi) **T**. Equal f-outputs give equal composite outputs, hence equal inputs.

(vii) **F**. Exactly one output per input is part of the function definition.

Marking: Award 1 point per correct T/F choice (7 total); reasons are included here for checking.

Problem 4 [16 points]

(a) **[2 points]** Arrange the four remaining cities between the fixed endpoints: $4!$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(b) **[2 points]** Choose the three image values, then use each at least once: $\binom{5}{3}(3^4 - 3 \cdot 2^4 + 3) = 360$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(c) **[2 points]** The zero-one-count cases are disjoint: one all-zero string plus eight single-one positions, giving $1 + 8 = 9$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(d) **[2 points]** After fixing the endpoints, the five free positions hold three reds and two blues: $5!/(3!2!) = 10$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(e) **[2 points]** Treat the prescribed triple as one circular unit. Five units give $(5 - 1)! = 4!$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(f) **[2 points]** Add the uniquely determined slack $w = 6 - x - y - z \geq 0$. The four-variable sum has $\binom{9}{3} = 84$ solutions.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(g) **[2 points]** If k factors contribute x^3, the exponent is $3k - 2(10 - k) = 5k - 20$. Thus $k = 4$, giving $\binom{10}{4}2^6 = 13440$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

(h) **[2 points]** Choose which of X, Y is outside in two ways, then choose its two outside companions from the other six files. Of the five files left inside, choose three for A; the other two go to B. Thus $2\binom{6}{2}\binom{5}{3} = 300$.
Marking: 1 mark for a valid counting setup; 1 mark for the correct exact answer. An equivalent exact expression earns full credit.

Problem 5 [10 points]

(a) **[6 points]** Fix an element a of an n-element set. Its r-element subsets split into those containing a and those not containing a. In the first class, choose the other $r-1$ elements from the remaining $n-1$, giving $\binom{n-1}{r-1}$. The second class has $\binom{n-1}{r}$ subsets. The classes are disjoint and exhaustive, so their sum is $\binom{n}{r}$.
Marking: 1 mark for fixing an element; 2 for each class count; 1 for the disjoint exhaustive conclusion.

(b) **[4 points]** For $a, b \geq 1$, the last step comes from $(a-1, b)$ or $(a, b-1)$, so $p(a,b) = p(a-1,b) + p(a,b-1)$. On either axis there is exactly one path, so $p(a,0) = p(0,b) = 1$. Unrestricted paths to $(4,3)$ total $\binom{7}{3} = 35$. Those through $(2,1)$ split uniquely into $\binom{3}{1}\binom{4}{2} = 18$ paths, leaving $35 - 18 = 17$.
Marking: 1 mark for the last-step recurrence; 1 for boundary values; 1 for total and forbidden-path counts; 1 for subtraction and answer.

Problem 6 [9 points]

[9 points] With an alphabet of size m, endpoint inequality gives $m(m-1)$ endpoint choices and m^4 internal choices, so $(m-1)m^5$ strings. Let E_A, E_B, E_C be the missing-symbol events. Missing a specified set of j required symbols leaves $m = 4 - j$. IE therefore gives

$$3 \cdot 4^5 - \binom{3}{1} 2 \cdot 3^5 + \binom{3}{2} 1 \cdot 2^5 - 0 \cdot 1^5 = 1710.$$

The triple intersection is empty because the one remaining symbol cannot make unequal endpoints.
Marking: 2 marks for the endpoint-constrained universe; 2 for the missing events; 3 for all intersection counts including the empty triple; 2 for the IE expression and answer.

Problem 7 [7 points]

[7 points] Map each string to its six-bit prefix; there are $2^6 = 64$ prefix boxes. Sixty-five strings force two with one prefix. Their strings are distinct, so their last bits must differ. For sharpness, take all 64 six-bit prefixes and append 0 to each. These strings have distinct prefixes, so no forced pair occurs.
Marking: 2 marks for the prefix map and box count; 2 for PHP; 1 for distinctness forcing the last-bit difference; 2 for the sharp construction.

Problem 8 [6 points]

(a) **[1 points]** The counts are $30, 20, 12$, so the probabilities are $1/2, 1/3, 1/5$.
Marking: 1 mark for all three probabilities.

(b) **[1 points]** Ten multiples of six give $P(A \cap B) = 1/6 = (1/2)(1/3)$; the events are independent.
Marking: 1 mark for the joint mass and independence comparison.

(c) **[1 points]** The counts are six multiples of ten and four multiples of fifteen, giving $1/10$ and $1/15$.
Marking: 1 mark for both joint probabilities.

(d) **[1 points]** Two multiples of thirty give $2/60 = 1/30$.
Marking: 1 mark for $1/30$.

(e) **[2 points]** The probability is

$$\frac{30 + 20 + 12 - 10 - 6 - 4 + 2}{60} = \frac{11}{15}.$$

Singletons are added, pair intersections subtracted, and the triple intersection restored.
Marking: 1 mark for correct inclusion-exclusion signs/counts; 1 for $11/15$.

Problem 9 [7 points]

(a) **[3 points]** Opening-2 likelihoods under prize doors $1, 2, 3$ are $1/2, 0, 1$. Hence $P(M = 2) = 1/4 + 1/6 = 5/12$. Staying wins with posterior $3/5$, and switching with $2/5$. Stay is better.
Marking: 1 mark for likelihoods and denominator; 1 for posteriors; 1 for the action.

(b) **[4 points]** $P(M = 3) = 1/4 + 1/3 = 7/12$. Staying wins with posterior $3/7$ and switching with $4/7$, so switch to door 2. The combined policy wins on the disjoint cases prize 1 and opening 2, or prize 2 and opening 3:

$$P(\text{win}) = (1/2)(1/2) + (1/3)(1) = 7/12.$$

Marking: 2 marks for opening-3 posteriors; 1 for switching; 1 for the overall joint-event sum.

Problem 10 [10 points]

(a) **[2 points]** $C = 0, 1$ each with probability $1/2$, so $E(C) = 1/2$ and $V(C) = 1/2 - (1/2)^2 = 1/4$.
Marking: 1 mark for the induced law; 1 for both moments.

(b) **[2 points]** For every $a, b \in \{0, 1\}$, $P(A = a, B = b) = 1/4 = (1/2)(1/2)$. All other joint masses and corresponding marginal products are zero. Thus the random variables are independent.
Marking: 1 mark for the four nonzero joint masses; 1 for factorization for all values.

(c) **[2 points]** For either pair the four binary joint values occur once each, with mass $1/4$, so both pairs are independent. But $P(A = 1, B = 1, C = 1) = 0 \neq 1/8$, so the three are not mutually independent.
Marking: 1 mark for both pairwise joint laws; 1 for the failing triple factorization.

(d) **[4 points]** Linearity gives $E(A + B - C) = 1/2$. Pairwise independence makes every cross-product expectation equal to the product of the corresponding means; this is exactly what the expansion of the centered square needs. Variance addition with squared coefficients gives

$$V(A + B - C) = 1/4 + 1/4 + (-1)^2/4 = 3/4.$$

The values on $(0, 0), (0, 1), (1, 0), (1, 1)$ are $0, 0, 0, 2$, verifying mean $1/2$ and variance $1 - 1/4 = 3/4$.
Marking: 1 mark for the mean; 1 for squared-coefficient variance; 2 for why pairwise factorization suffices (or an explicit centered-square derivation).